

9.7.1

EE25BTECH11057 - Rushil Shanmukha Srinivas

Question : Find the area of the region $\{S(x, y) : x^2 + y^2 \leq 16a^2 \text{ and } y^2 \leq 6ax\}$.

Solution :

Name	Value
Parabola	$\mathbf{x}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - (6a \ 0) \mathbf{x} = 0$
Circle	$\mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - 16a^2 = 0$

Table : Conics

On comparing conics with

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (0.1)$$

The parameters of the Parabola are

$$\mathbf{V}_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u}_1 = -\begin{pmatrix} 3a \\ 0 \end{pmatrix}, f_1 = 0 \quad (0.2)$$

The parameters of the Circle are

$$\mathbf{V}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u}_2 = 0, f_2 = -16a^2 \quad (0.3)$$

The intersection of two conics with parameters $\mathbf{V}_i, \mathbf{u}_i, f_i$, $i = 1, 2$ is defined as

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0. \quad (0.4)$$

(0.1) represents a pair of straight lines if

$$\begin{vmatrix} \mathbf{V} & \mathbf{u} \\ \mathbf{u}^\top & f \end{vmatrix} = 0 \quad (0.5)$$

From (0.5), (0.4) represents a pair of straight lines if

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \quad (0.6)$$

On substituting from (0.2) and (0.3) in (0.6), we obtain

$$\begin{vmatrix} \mu & 0 & -3a \\ 0 & \mu + 1 & 0 \\ -3a & 0 & -16\mu a^2 \end{vmatrix} = 0 \quad (0.7)$$

On solving we get

$$(\mu + 1)(16\mu^2 + 9) = 0 \quad (0.8)$$

yielding

$$\mu = -1 \quad (0.9)$$

Substituting (0.2) and (0.3) in (0.4), we obtain

$$\mathbf{x}^\top \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} -3a & 0 \end{pmatrix} \mathbf{x} + 16a^2 = 0 \quad (0.10)$$

On solving we get

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 2a \text{ and } \begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = -8a \quad (0.11)$$

But circle and line $\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = -8a$ do not intersect. So The solution is

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 2a \quad (0.12)$$

Therefore the intersection of two conics is a line with parameters

$$\mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \mathbf{h} = \begin{pmatrix} 2a \\ 0 \end{pmatrix} \quad (0.13)$$

The points of intersection for a conic and a line is given by

$$\kappa_i = \frac{1}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \left(-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})} \right) \quad (0.14)$$

$$g(\mathbf{x}) = \mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - 16a^2 \quad (0.15)$$

$$g(\mathbf{h}) = \mathbf{h}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h} - 16a^2 \quad (0.16)$$

The intersection parameters of the chord in (0.12) with the circle is obtained as

$$\kappa_i = \pm \sqrt{16a^2 - \mathbf{h}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h}} \quad (0.17)$$

$$\kappa_i = \pm 2\sqrt{3}a \quad (0.18)$$

So the points of intersection are obtained as

$$\mathbf{x}_1 = \begin{pmatrix} 2a \\ 2\sqrt{3}a \end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix} 2a \\ -2\sqrt{3}a \end{pmatrix} \quad (0.19)$$

The desired area of region is given as

$$2 \left(\int_0^{2a} \sqrt{6ax} dx + \int_{2a}^{4a} \sqrt{16a^2 - x^2} dx \right) \quad (0.20)$$

$$= 2 \left[\sqrt{6a} \left(\frac{2}{3} x^{3/2} \right) \right]_0^{2a} + 2 \left[\frac{x}{2} \sqrt{16a^2 - x^2} + 8a^2 \sin^{-1} \left(\frac{x}{4a} \right) \right]_{2a}^{4a} \quad (0.21)$$

$$= \frac{4a^2}{\sqrt{3}} + \frac{16\pi a^2}{3} \quad (0.22)$$

$$= 4a^2 \left(\frac{4\pi}{3} + \frac{1}{\sqrt{3}} \right) \quad (0.23)$$

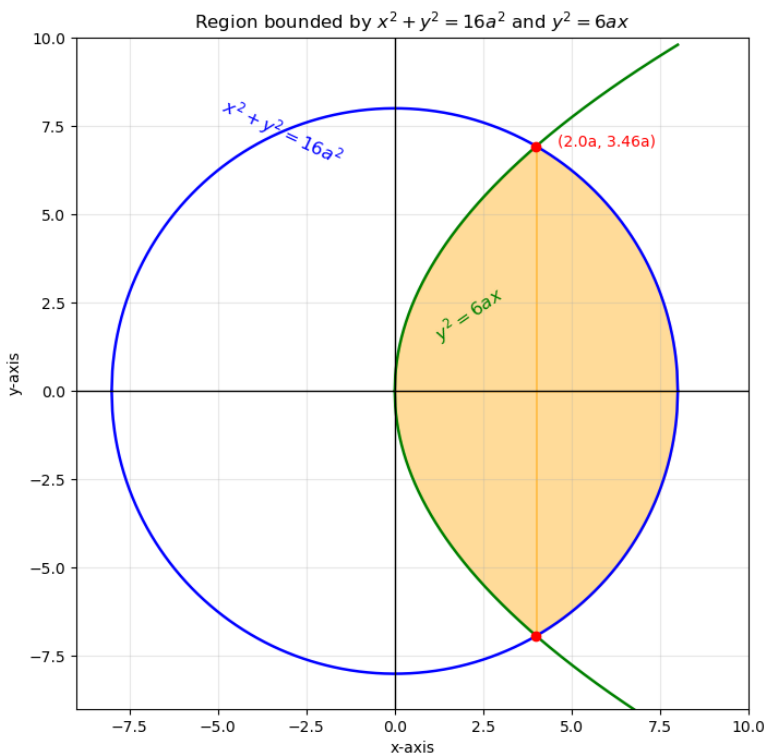


Fig : Parabola and Circle